

Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
- Write all answers sequentially.
- Start answering a question (not the part of the question) from the top of a new page.
- Write legibly and in orderly fashion maintaining all mathematical norms and rules. Prepare a single solution file.
- Start working right away. There is no late submission form. If you miss the deadline, you need to use the make-up assignment to cover up the marks.

The floating point representation can be expressed in any of the following forms:

Standard Form	:	$F = (\pm 0.d_1d_2d_3 \dots d_m)_\beta \times \beta^e; \quad d_1 \neq 0.$
IEEE Normalized Form	:	$F = (\pm 0.1d_1d_2d_3 \dots d_m)_\beta \times \beta^e.$
IEEE Denormalized Form	:	$F = (\pm 1.d_1d_2d_3 \dots d_m)_\beta \times \beta^e.$

where $d_i, \beta, e \in \mathbb{Z}$, $0 \leq d_i \leq \beta - 1$ and $e_{\min} \leq e \leq e_{\max}$.

Question: 1 (10 Marks) (Graded Part)

Given the real number $x = (3.165)_{10}$.

- Convert x into its binary representation, ensuring at least 9 places after the binary/decimal point. [3 Marks]
- Determine the binary representation of x [Find $fl(x)$] when stored in a floating-point system using the standard form, considering mantissa sizes of $m = 4$ and $m = 6$. [4 Marks]
- Convert the stored binary values back to decimal form and compute the **rounding error** for each case. [3 Marks]

Question: 2 (10 Marks) (Graded Part)

Now, consider a system with $\beta = 2$, $m = 4$, and $-1 \leq e \leq 1$.

- Find the **maximum** and **minimum** numbers that can be stored in **IEEE Normalized Form**, and **IEEE Denormalized Form** [with and without negative support]. Express these numbers in both binary and decimal formats. [3 Marks]
- How many numbers can this system represent or store in all these forms? [2 Marks]
- Now, consider $m = 3$. Using **Standard Form**, find all the decimal numbers without **negative support**, plot them on a real line, and show if the number line is **equally spaced or not**. [5 Marks]

Question: 3 (10 Marks) (Ungraded Part)

Consider the following function, $v(t) = -\frac{3}{4}t^2 + \frac{19}{2}t - 6$, and $t_0 = 2$, $t_1 = 4$, $t_2 = 6$. Based on these, answer the following questions:

- Find an interpolating polynomial for velocity that passes through the given data points using the Vandermonde Matrix method. Additionally, determine an approximate acceleration value at $t = 7$ seconds. [4+1 marks]
- Determine an interpolating polynomial for velocity that passes through the given data points using the Lagrange method. [5 marks]

Question: 4 (10 Marks) (Ungraded Part)

Suppose, a toy robot uses a floating-point system $\beta = 2$, $m=5$, $-200 \leq e \leq 200$ to measure distances in a game. Now, answer the following questions:

- (a) Compute the **maximum delta value** for the **standard form**. [2 Marks]
- (b) Compute the minimum value of $|x|$ for **normalized** and **denormalized** forms. [4 Marks]
- (c) Compute the **machine epsilon** for the **normalized** and **denormalized** forms. [4 Marks]

Question: 5 (10 Marks) (Ungraded Part)

Read the following questions and answer accordingly:

1. Given the nodes $\left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$, determine an interpolating polynomial of the appropriate degree using Newton's divided difference method for the function $f(x) = x \sin(x)$. [4 marks]
2. Utilize the interpolating polynomial to estimate the value at $\frac{\pi}{4}$, and determine the percentage relative error at $\frac{\pi}{4}$. [2 marks]
3. Insert a new node π into the given set of nodes and determine the interpolating polynomial of the appropriate degree. [4 Marks]